

A Separable Banach-Space Counterexample to Strong Approximation by Two Idempotents

fa_banach_001, agent_lane_01

August 26, 2026

Status

Candidate counterexample, likely valid, pending human review.

Source Question

Jean-Christophe Bourin, *A Journey into Matrix Analysis*, arXiv:2307.03064, Question 5.2.9 on PDF page 82, asks:

Let $\mathcal{X}$ be a separable Banach space and $T \in \mathcal{L}(\mathcal{X})$. Do there exist two sequences $\{P_n\}_{n=1}^\infty$ and $\{Q_n\}_{n=1}^\infty$ of idempotents in $\mathcal{L}(\mathcal{X})$ such that

$$T = \text{sot} \lim_{n \rightarrow \infty} (P_n + Q_n)?$$

82

CHAPTER 5. PINCHINGS AND MASAS

It is known [85] that any operator is the sum of five idempotents. We close this section by asking whether Corollary 5.2.5 admits a substitute for Banach space operators.

Question 5.2.9. Let $\mathcal{X}$ be a separable Banach space and $T \in \mathcal{L}(\mathcal{X})$, the linear operators on $\mathcal{X}$. Do there exist two sequences $\{P_n\}_{n=1}^\infty$ and $\{Q_n\}_{n=1}^\infty$ of idempotents in $\mathcal{L}(\mathcal{X})$ such that $T = \text{sot} \lim_{n \rightarrow +\infty} (P_n + Q_n)$?

5.3 Conditional expectation onto a masa

5.3.1 Conditional expectation of general operators

Let $\mathcal{M}$ be a von Neumann algebra acting on a Hilbert space $\mathcal{H}$. Let $\mathcal{A}$ be a maximal abelian self-adjoint algebra (masa) in $\mathcal{M}$. Let $T \in \mathcal{M}$. The conditional expectation of T onto $\mathcal{A}$ is the unique operator $E_{\mathcal{A}}(T) \in \mathcal{A}$ such that

Here strong-operator convergence means pointwise norm convergence on $\mathcal{X}$.

Idea of the Proof

The Hilbert-space construction behind the source question uses the abundance of large complemented subspaces. The obstruction is to choose a separable space at the opposite extreme. On an indecomposable Banach space, every idempotent has finite-dimensional range or finite-dimensional kernel. Thus a sum of two idempotents is always a finite-rank perturbation of exactly one of 0 , I , or $2I$.

Choose an indecomposable separable space without the compact approximation property. If sums of two idempotents converged strongly to $\frac{1}{2}I$, one of the three scalar types would occur along an infinite subsequence. Removing that fixed scalar gives finite-rank operators converging strongly to a nonzero scalar multiple of the identity. Uniform boundedness then turns them into bounded finite-rank approximants to I , which converge uniformly on compact sets. This is the bounded approximation property and contradicts the choice of the space.

Counterexample

Lemma 1. *Let X be an indecomposable infinite-dimensional Banach space. For every idempotent $P \in \mathcal{L}(X)$ there are $\varepsilon \in \{0, 1\}$ and a finite-rank operator F such that*

$$P = \varepsilon I + F.$$

Proof. Since $P^2 = P$, both $P[X]$ and $\ker P$ are closed and

$$X = P[X] \oplus \ker P.$$

Indecomposability forces one of these two summands to be finite-dimensional. If $P[X]$ is finite-dimensional, take $\varepsilon = 0$ and $F = P$. If $\ker P$ is finite-dimensional, then $I - P$ is finite-rank, so take $\varepsilon = 1$ and $F = P - I$. $\square$

Lemma 2. *Suppose finite-rank operators $R_j \in \mathcal{L}(X)$ satisfy $R_j x \rightarrow x$ for every $x \in X$ and $\sup_j \|R_j\| < \infty$. Then X has the bounded approximation property, and therefore the compact approximation property.*

Proof. Write $M = \sup_j \|R_j\|$. Let $K \subset X$ be compact and $\eta > 0$. Choose a finite $\eta/(3(M+1))$ -net $\{x_1, \dots, x_m\}$ in K . For all sufficiently large j , $\max_i \|R_j x_i - x_i\| < \eta/3$. If $x \in K$ and $\|x - x_i\| < \eta/(3(M+1))$, then

$$\|R_j x - x\| \leq M\|x - x_i\| + \|R_j x_i - x_i\| + \|x_i - x\| < \eta.$$

Hence $R_j \rightarrow I$ uniformly on compact subsets, with uniformly bounded norms. This is the bounded approximation property. Since finite-rank operators are compact, it implies the compact approximation property. $\square$

Theorem 1. *There exists a separable infinite-dimensional Banach space X for which $\frac{1}{2}I_X$ is not a strong-operator limit of sums of two idempotents. Consequently Question 5.2.9 of arXiv:2307.03064 has a negative answer.*

More generally, on the chosen space, λI_X is not such a limit for any scalar $\lambda \notin \{0, 1, 2\}$.

Proof. Alexandrov, Kutzarova, and Plichko prove that there is a closed subspace X of the Gowers–Maurey hereditarily indecomposable space which fails the compact approximation property [2]. In particular, X is separable, infinite-dimensional, and indecomposable.

Fix $\lambda \notin \{0, 1, 2\}$ and suppose, toward a contradiction, that idempotents $P_n, Q_n \in \mathcal{L}(X)$ satisfy

$$P_n + Q_n \xrightarrow{\text{st}} \lambda I.$$

By Lemma 1, write

$$P_n = \varepsilon_n I + F_n, \quad Q_n = \delta_n I + G_n,$$

where $\varepsilon_n, \delta_n \in \{0, 1\}$ and F_n, G_n are finite-rank. Put $k_n = \varepsilon_n + \delta_n \in \{0, 1, 2\}$. One value $k \in \{0, 1, 2\}$ occurs along an infinite subsequence (n_j) . Along this subsequence,

$$F_{n_j} + G_{n_j} \xrightarrow{\text{st}} (\lambda - k)I.$$

The uniform boundedness principle gives $\sup_j \|F_{n_j} + G_{n_j}\| < \infty$. Because $\lambda - k \neq 0$, the finite-rank operators

$$R_j = (\lambda - k)^{-1}(F_{n_j} + G_{n_j})$$

are uniformly bounded and converge strongly to I . Lemma 2 then gives the compact approximation property for X , contradicting its construction. Taking $\lambda = \frac{1}{2}$ gives the announced counterexample. $\square$

Verification Notes

No bound on the individual idempotents is assumed or needed. Strong convergence bounds the sums $P_n + Q_n$; after the scalar type is fixed on a subsequence, it therefore bounds the finite-rank remainders used in the proof. The changing scalar types cause no difficulty because there are only three of them. The distinction between pointwise convergence and uniform convergence on compact sets is handled explicitly in Lemma 2.

Novelty Check and Limitations

The local run indexes were searched for arXiv:2307.03064, the source title, the exact question wording, strong limits and sums of two idempotents, and the bounded/compact approximation properties. No matching packet or attempt was found. The surrounding source section was checked and does not answer Question 5.2.9.

A bounded web search on August 26, 2026 using the exact question and close strong-limit phrases found no later answer or this approximation-property obstruction. This was not a comprehensive MathSciNet or zbMATH search, so novelty confidence is moderate.

The argument disproves the universal Banach-space statement. It does not classify the separable Banach spaces on which every operator has such an approximation, nor does it weaken the HI/CAP construction to a familiar classical space.

Human Review Recommendation

Send to a Banach-space expert. The proof itself is elementary once the supporting existence proposition is accepted; the main review item is the scalar-field scope of that standard HI construction.

References

- [1] J.-C. Bourin, *A Journey into Matrix Analysis*, arXiv:2307.03064, 2023.
- [2] G. Alexandrov, D. Kutzarova, and A. Plichko, *A Separable Space with No Schauder Decomposition*, Proceedings of the American Mathematical Society 127 (1999), no. 9, 2805–2806, doi:10.1090/S0002-9939-99-05370-8.